

ABHINAV JHA

Computer Science & Engineering Student | Aspiring GenAI Engineer

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PROFESSIONAL SUMMARY

Final-year Computer Science Engineering student with hands-on experience building applications powered by Large Language Models and generative AI, including a FastAPI-based retrieval-augmented generation (RAG) service using a Groq-hosted LLM. Comfortable working with modern AI tools (ChatGPT, Gemini, Claude) and experienced in prompt engineering, embeddings, and LLM API integration through self-built projects. Strong analytical foundation from machine learning and computer vision projects, with working knowledge of Python backend frameworks such as FastAPI and LLM tooling such as LangChain. Looking to grow as an AI/GenAI Engineer, contributing to real-world LLM-powered systems under industry mentorship.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

AI / GenAI: Generative AI, LLM Engineering, RAG, Prompt Engineering, Embeddings, NLP, Machine Learning, Deep Learning, Computer Vision, YOLO

GenAI Tools: ChatGPT, Google Gemini, Claude, Groq API

Frameworks: LangChain, FastAPI, PyTorch, Scikit-Learn, TensorFlow, Keras, Hugging Face

Web Technologies: HTML, CSS, JavaScript

Databases: MySQL, Supabase

Tools & Platforms: Git, Streamlit, AWS

PROJECT EXPERIENCE (AI / GENAI)

AI Q&A Chatbot (LLM-based)

- Built a conversational Q&A chatbot using a large language model API, deployed through an interactive Streamlit interface for real-time use.
- Applied prompt engineering techniques to design context-aware prompt handling, improving the accuracy and relevance of responses to user queries.

YouTube Video Q&A — Chrome Extension + Local Backend (RAG)

- Built a Chrome extension that injects a floating Q&A panel onto YouTube watch pages, letting users ask questions about the video they are currently watching and get answers generated from its full transcript.
- Built a FastAPI backend implementing a retrieval-augmented generation (RAG) pipeline: fetched transcripts (youtube-transcript-api, with a yt-dlp fallback), chunked and embedded the text, and answered queries using a Groq-hosted LLM.
- Designed the system so the extension never calls an AI API directly — all transcript retrieval, embedding, and LLM calls run through the local backend service, keeping the architecture modular and API keys secure.

Driver Drowsiness Detection System (ML + Computer Vision)

- Developed a real-time driver monitoring system using computer vision and deep learning (YOLO) to detect eye closure and fatigue patterns.
- Implemented OpenCV-based video processing and optimized model inference for real-time performance, ensuring accurate detection under varying lighting and driving conditions.
- Created a custom dataset for the project and built an instant-alert feature to reduce accident risk.

EDUCATION

B.Tech in Computer Science & Engineering — VIT Bhopal University | CGPA: 8.54 | Expected 2027

Science Stream — Jesus Mary & Academy | 2022

CERTIFICATIONS

- Applied Machine Learning in Python — University of Michigan
- Oracle Cloud Infrastructure Certified AI Foundations Associate
- TCS iON Career Edge - AI Foundation

LANGUAGES

English, Hindi